MYBHARAT HACKATHON PROJECT PROPOSAL

SMART HOSPITAL INDOOR NAVIGATION SYSTEM

Project vision
Helping patients and visitors find the right place inside a hospital with less confusion, more independence, and a more reassuring experience.

Project category: Digital Healthcare • Accessibility • AI-enabled Public Service

Proposed platform: Android application + responsive website + hospital administration dashboard

Hackathon focus: a practical, inclusive, scalable solution that can begin with a low-cost QR-based prototype and expand to indoor positioning and AI assistance.

Prepared as a project concept and implementation plan. Final deployment requires verified hospital maps and approval from the participating hospital.

1. Executive Summary

The Smart Hospital Indoor Navigation System is a digital wayfinding solution designed to help patients, visitors, caregivers, and hospital staff navigate complex hospital buildings. Users can search for a department, diagnostic lab, pharmacy, ward, room number, lift, or other facility and receive clear directions on a 2D indoor map.

The proposed solution combines QR-code-based starting locations, floor-by-floor maps, accessible route options, multilingual instructions, voice and vibration guidance, and an optional Bluetooth Low Energy (BLE) positioning module. An administration dashboard allows authorised hospital staff to maintain maps, rooms, routes, QR codes, and translations. A Super Admin can manage multiple hospitals.

The first prototype should focus on one verified floor and QR-based navigation. BLE real-time positioning and AI assistance can be added after the core navigation has been tested.

2. Problem Statement

Large hospitals often contain multiple buildings, floors, corridors, departments, and diagnostic services. First-time visitors may not know where to go. Patients may be anxious, older adults may struggle with complex signs, and families may be worried while trying to reach a ward or treatment area. Staff can also spend time repeatedly giving directions.

- Visitors can find it difficult to identify the correct department, room, or floor.
- Language differences can make signs and verbal directions harder to understand.
- People with mobility needs may require routes that avoid stairs and use accessible lifts or ramps.
- Room changes or closed corridors can make printed directions outdated.
- A digital solution must still support people who do not have a smartphone or cannot use an app.

3. Proposed Solution

Build a simple, accessible hospital navigation platform for Android and the web. Visitors can scan a QR code at reception or a known location, choose their destination, and follow a route on the hospital's verified indoor map. The platform can provide step-by-step text, voice prompts, vibration cues, language selection, and wheelchair-friendly route preferences.

- Interactive 2D maps for each approved floor.
- Search for departments, services, facilities, and room numbers.
- QR scanning to identify a known starting point.
- Optional BLE beacon integration for estimated real-time indoor positioning.
- Accessible route preferences for lifts, ramps, and step-free paths.
- Multilingual support: English, Tamil, Malayalam, Telugu, Kannada, Hindi, with optional Tulu.
- Admin dashboard for hospital maps, routes, QR codes, translations, and access permissions.
- AI assistant that matches user questions to approved hospital locations, with a human-help fallback.

4. Project Objectives

- Make hospital navigation easier for first-time visitors.
- Reduce avoidable confusion and unnecessary wandering.
- Support independent navigation while keeping reception assistance visible.
- Improve access for people who prefer local-language, audio, or step-free guidance.
- Provide hospital staff with a maintainable, centrally managed map system.
- Create a scalable model that can be adapted to multiple hospitals.

5. User Journey

Step	User action	System response
1. Start	Open the app/site or scan a QR code.	Loads the hospital and starting location.
2. Search	Choose a destination such as MRI, pharmacy, or Room 205.	Shows verified matching locations.
3. Route	Select the correct destination and route preference.	Calculates a path on the approved map graph.
4. Navigate	Follow the map and step-by-step directions.	Provides floor changes and optional voice/vibration cues.
5. Finish	Reach the destination or request help.	Shows arrival confirmation or reception fallback.

6. Main Features and Modules

Module	Key functions
Visitor app and website	Search, map view, destination selection, directions, language settings, help contact.
2D indoor map	Floor plans, room markers, corridors, lifts, stairs, ramps, entrances, restricted areas.
QR navigation	Unique location QR codes, scan validation, known starting point, QR status management.
BLE positioning (optional)	Beacon records, estimated current position, calibration, uncertainty indication, maintenance.
Accessibility	Voice directions, vibration cues, larger text, high contrast, step-free route preference.
AI assistant	Natural-language destination search, clarification when results are ambiguous, approved translations.
Hospital admin	Edit approved maps, rooms, routes, QR codes, language content, and maintenance status.
Super Admin	Onboard hospitals, assign administrators, manage platform-wide settings and access.
Analytics	Aggregated destination searches, route completion, common help requests, map update status.

7. Advantages and Expected Impact

Advantage	Potential value
Less confusion	Visitors can search for a destination instead of guessing where to go.
Time efficiency	Clear routes may reduce detours; actual improvements must be measured in a pilot.
Better visitor experience	Simple guidance can make unfamiliar hospital spaces feel more manageable.
More independence	Visitors may need fewer repeated directions from staff.
Accessibility	Audio, vibration, readable text, and verified step-free routes support more users.
Language inclusion	Users can receive directions in a familiar language.
Maintainability	Authorised staff can update room details and temporarily close routes centrally.
Scalability	The same platform design can support multiple buildings and hospitals.

8. Emotional and Human Impact

A hospital visit can come with fear, uncertainty, pain, or concern for a loved one. Searching for the correct room should not add unnecessary stress to an already difficult day. Clear navigation cannot remove the emotional burden of illness, but it can reduce one avoidable source of confusion and help people feel more confident about their next step.

Emotional design principles

- Reassurance: use calm screens, simple language, and clear next steps.
- Dignity: make it easier to find a place without feeling embarrassed about asking for directions.
- Independence: help people navigate by themselves while offering human support when needed.
- Family support: make it easier for visitors to reach the right ward or service.
- Trust: be honest when location accuracy is uncertain; never pretend a route is guaranteed.

- Inclusion: consider language, age, disability, digital access, and different levels of technology familiarity.

Suggested emotional message: “A hospital visit can already bring many worries. No one should have to feel lost while trying to find care for themselves or someone they love. Our goal is to make every direction clearer, every step more reassuring, and every visitor feel more confident inside the hospital.”

9. Technology Stack

Layer	Suggested technology	Purpose
Android app	Flutter	Mobile interface and navigation experience.
Website	React / Next.js	Responsive visitor site and admin dashboard.
Backend	Python FastAPI or Node.js	Search, route services, permissions, and content management.
Database	PostgreSQL	Hospitals, floors, locations, paths, users, and audit logs.
Indoor maps	SVG-based 2D maps	Scalable map display and location markers.
Route engine	A* or Dijkstra algorithm	Path calculation using verified walkable connections.
Authentication	Firebase Auth or approved OTP provider	Secure admin and optional user login.
AI integration	Gemini API or another approved LLM	Natural-language destination assistance.
Hosting	Suitable cloud hosting / free tier for demo	Deploy API, website, and map assets.

10. Basic System Architecture

Visitor App / Website → Backend API → Hospital Directory + Route Engine → Database and Map Storage. The optional AI assistant translates natural-language requests into searches against approved locations. Optional BLE services estimate location using deployed and calibrated beacons. The admin dashboard updates maps and route data through permission-controlled APIs.

A route graph uses nodes for corridor junctions, room entrances, lifts, and stairs. Edges represent verified walkable connections. A* or Dijkstra calculates a route. For wheelchair routing, the graph must exclude stairs and include only verified accessible lifts, ramps, and paths.

11. Database Outline

Table	Example fields
hospitals	id, name, address, logo, status
buildings	id, hospital_id, name
floors	id, building_id, floor_number, map_asset, reviewed_at
locations	id, floor_id, name, category, room_number, coordinates
path_nodes / path_edges	node IDs, connected nodes, distance, accessibility, route status
qr_codes	id, location_id, token, status, last_checked
beacons	id, floor_id, location_id, beacon_identifier, status
users_roles	user_id, hospital_id, role, permissions
translations	content_key, language_code, translated_text, review status
audit_logs	actor, action, timestamp, affected record

12. Security, Privacy, and Safety

- Use HTTPS, secure authentication, and role-based access control.
- Keep each hospital's data isolated from other hospitals.
- Collect the minimum data required; avoid unnecessary patient records or continuous location histories.
- Explain location permissions clearly and make tracking optional where practical.
- Use audit logs, backups, access reviews, and a defined data-retention policy.
- Require hospital staff to verify maps, routes, accessible paths, and temporary closures.
- Do not replace official emergency signage, fire-safety procedures, or staff instructions.
- If a destination or route is unknown, direct the user to reception rather than guessing.

13. Development Plan

Phase	Activities	Deliverable
Phase 1: Define	Choose one pilot hospital; obtain approved floor plans and location list.	Agreed scope and verified source map.
Phase 2: Map	Create a 2D map, nodes, corridors, floor links, and destination directory.	One testable floor map.
Phase 3: Prototype	Build search, QR scan, route calculation, and step-by-step directions.	QR-based navigation MVP.
Phase 4: Admin	Add map and location editing, QR management, roles, and audit trail.	Basic management portal.
Phase 5: Accessibility	Add languages, voice, vibration, and step-free route option.	Inclusive navigation experience.
Phase 6: Pilot test	Test with visitors and staff; fix map and usability issues.	Validated pilot report.
Phase 7: Expand	Add BLE positioning, AI assistant, and more hospitals as needed.	Advanced multi-hospital version.

14. Testing and Success Metrics

- Route completion rate: percentage of test journeys that reach the correct destination.
- Time to destination: compare observed journey times in controlled pilot tests.
- Wrong-turn and help-request rate: record common points of confusion.
- Accessibility success: test voice, text scaling, and step-free routes with representative users.
- Language usability: verify that instructions are understandable in each supported language.
- Map freshness: track how quickly verified changes are published.
- User confidence: use a short voluntary survey about clarity, confidence, and stress.

15. Risks and Mitigation

Risk	Mitigation
Outdated map	Assign a map owner and review changes with hospital facilities staff.
QR damaged or missing	Inspect signs regularly and keep reception help available.
BLE location inaccurate	Calibrate on site, show uncertainty, and test across devices.
Network failure	Cache essential map information where possible and provide physical signage.
Translation error	Have qualified hospital staff review important directions.
Digital exclusion	Keep non-digital signs and human assistance available.

16. Budget and Sustainability

Indicative estimates only; these are not supplier quotations and exclude some costs such as staff time, map surveying, taxes, and ongoing support.

Prototype / deployment	Indicative estimate (INR)	Notes
Basic demo	₹0–₹5,000	One sample floor and simple QR workflow; own time excluded.
QR-based MVP	₹5,000–₹30,000	Small prototype; final cost depends on scope and tools.
Multi-floor pilot	₹30,000–₹2 lakh	More mapping, admin functions, and testing.
BLE pilot	₹1 lakh–₹5 lakh+	Hardware, installation, calibration, and maintenance.
Production platform	₹5 lakh–₹25 lakh+	Broad estimate; security, integrations, support, and scale affect cost.

Potential sustainability model: hospital onboarding fees, annual subscriptions, map maintenance, QR signage services, BLE installation and support, and enterprise licensing for hospital groups. A pilot should first demonstrate usefulness and reliability before making claims about savings.

17. Innovation and Differentiation

- QR-first approach makes a low-cost starting prototype possible.
- Multi-hospital architecture supports future scaling through one Super Admin platform.
- Local-language, voice, and vibration directions focus on practical inclusion.
- Accessible routing is part of the route engine, not merely a visual setting.
- AI is constrained to approved location data, reducing the risk of invented directions.
- The design keeps human assistance available instead of assuming every visitor can use a smartphone.

18. Hackathon Pitch Summary

Problem: Large hospitals can be confusing to navigate, especially for first-time visitors, older adults, people with disabilities, and worried families.

Solution: An indoor navigation app and website with 2D maps, QR-based starting points, accessible routes, multilingual guidance, and an admin dashboard, with BLE and AI as optional extensions.

Impact: The project aims to reduce avoidable confusion, support independence, and make hospital spaces easier to navigate. The impact should be demonstrated through pilot testing and user feedback.

Feasibility: Start with one hospital floor and QR navigation, validate it with real users, then expand to BLE positioning and multiple hospitals.

Closing statement: “This project is more than a map. It is a way to help people feel less lost at a moment when they may already be worried. By making hospital directions clearer and more inclusive, we can make the journey to care easier to understand.”

19. AI App-Builder Prompt

Build a responsive Android app and website called Smart Hospital Indoor Navigation System. Include a visitor interface, a hospital admin dashboard, and a Super Admin portal for multiple hospitals. Implement a searchable directory of departments, services, rooms, and facilities; interactive 2D floor maps; QR scanning to set a known starting point; destination selection; and route calculation using a verified walkable-path graph. Add floor switching, step-by-step text directions, multilingual UI, voice and vibration options, and a wheelchair-accessible route preference. Use Flutter for Android, React/Next.js for web, FastAPI or Node.js for backend, PostgreSQL for data, SVG maps, and A* or Dijkstra for routing. Include admin functions for hospitals, buildings, floors, locations, paths, QR codes, translations, user roles, and audit logs. Keep each hospital's data isolated. Add optional BLE beacon positioning and an AI assistant that searches only approved location data. The AI must not invent rooms, routes, medical advice, or emergency directions. Use sample data clearly labelled as sample until approved floor plans are supplied. Include security controls, privacy minimisation, accessibility testing, a test plan, and deployment instructions. Clearly label features that are prototypes or not yet connected; do not claim real-time tracking or production readiness unless tested.

Final Recommendation

For the hackathon, demonstrate a small working prototype rather than only a large feature list: one sample floor map, several destinations, QR-based starting points, route calculation, language selection, and an admin update flow. Explain how the prototype can grow into BLE and AI features. Validate the concept with a few representative users and present measured results honestly.